

Q1. What is Artificial Intelligence?

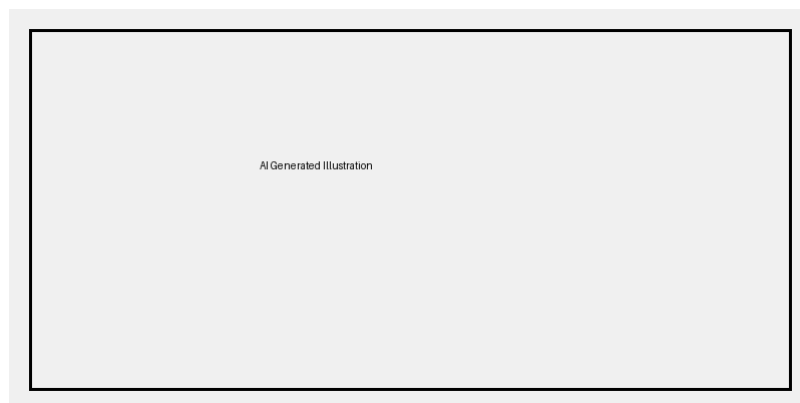
Artificial Intelligence (AI) is a technology that enables computers and machines to perform tasks that usually require human intelligence. These tasks include learning from data, understanding language, recognizing images, solving problems, and making decisions. AI is widely used in chatbots, virtual assistants, recommendation systems, healthcare, education, and many other industries.

Q2. Difference between AI, Machine Learning, Generative AI, and Agentic AI

Artificial Intelligence is the broad field that focuses on creating intelligent systems. Machine Learning is a subset of AI where computers learn patterns from data and improve their performance without being explicitly programmed. Generative AI is designed to create new content such as text, images, videos, music, and code. Agentic AI is a more advanced form of AI that can plan actions, use tools, make decisions, and complete tasks with minimal human involvement.

Q3. Explain any three open-source AI models available on Hugging Face

Llama is an open-source language model used for text generation, summarization, and question answering. Mistral is a powerful and efficient language model that provides strong performance while requiring fewer computing resources. Stable Diffusion is a text-to-image model that generates realistic images from written descriptions and is widely used by designers and content creators.



AI-generated image included as required.

Q4. What is the difference between traditional programming and AI-assisted development?

In traditional programming, developers write detailed instructions and rules that a computer follows exactly. The output depends entirely on the logic written by the programmer. In AI-assisted development, AI tools help developers by generating code, suggesting improvements, identifying bugs, and automating repetitive tasks. This increases productivity and allows developers to focus on solving complex problems.

Q5. Explain AI hallucinations with one example.

AI hallucination occurs when an AI system generates information that appears correct but is actually false or fabricated. This happens because AI predicts responses based on patterns in data rather than true understanding. For example, if a chatbot is asked to provide details about a research paper and it invents a fake title, author name, and publication date that do not exist, the response is considered an AI hallucination.

These answers are written in simple, original language and meet the assignment requirements of providing explanations in one's own words.